Chapter 10

Descriptive Statistics

10.1 Introduction

Statistics is concerned with the scientific method by which information is collected, organised, analysed and interpreted for the purpose of description and decision making.

Examples of firms using statistics are; RTSA, ZRA, CSO, Insurance companies etc.

There are two subdivisions of statistical methods.

(a) **Descriptive Statistics** - It deals with the presentation of numerical facts, or data, in either tables or graphs, and with the methodology of analysing the data.

(b) **Inferential Statistics** - It involves techniques for making inferences about the whole population on the basis of observations obtained from samples.

10.2 Some Basic Definitions

Definition 10.2.1. Population

A population is the group from which data are to be collected. For example, Cars, Animals, Trees, etc.

Definition 10.2.2. Sample

A sample is a subset of a population. For example, Saloons, Cows, Mukula trees, etc.

Definition 10.2.3. Discrete Data

A discrete data is one which no value may be assumed between two given values, for example, number of children in a family.

Discrete data are countable in a finite amount of time. Thus, discrete data takes only exact values. Cars, tomatoes etc.

Definition 10.2.4. Continuous Data

A continuous data is one which any value may be assumed between two given values.

Continuous data cannot take exact values but can be given only within a speficified range or measured to a specified degree of accuracy. For example, the time taken for 100-meter race, height, speed of a vehicle etc. run.

10.3 Frequency Distribution

Statistical data obtained by means of census, sample surveys or experiments usually consist of raw, unorganized sets of numerical values. Before these data can be used as a basis for inferences about the phenomenon under investigation or as a basis for decision making, they must be summarized and the pertinent information must be extracted.

A traffic inspector has counted the number of automobiles passing a certain point in 100 successive 20-minute time periods. The observations are listed below.

23	20	16	18	30	22	26	15	5	18
14	17	11	37	21	6	10	20	22	25
19	19	19	20	12	23	24	17	18	16
27	16	28	26	15	29	19	35	20	17
12	30	21	22	20	15	18	16	23	24
15	24	28	19	24	22	17	19	8	18
17	18	23	21	25	19	20	22	21	21
16	20	19	11	23	17	23	13	17	26
26	14	15	16	27	18	21	24	33	20
21	27	18	22	17	20	14	21	22	19

A useful method for summarizing a set of such data is the construction of a frequency table, or a frequency distribution. That is, we divide the overall range of values into a number of classes and count the number of observations that fall into each of these classes or intervals.

The general rules for constructing a frequency distribution table are;

i) There should not be too few or too many classes.

ii) Generally speaking, equal class intervals are preferred. But the first and last classes can be open-ended to cater for extreme values.

iii) Each class should have a class mark to represent the classes. It is also named as the class midpoint of the ith class. It can be found by taking simple average of the class boundaries or the class limits of the same class.

1. Setting up the classes for the above data.

   Choose a class width of 5 for each class, then we have seven classes going from 5 to 9, from 10 to 14, â€ę, and from 35 to 39.

2. Tallying and counting.

Classes	Tally Marks	Count
5-9		3
10 â€Ş 14		9
15 â€Ş 19		36
20 â€Ş 24		35
25 â€Ş 29		12
30 â€Ş 34		3
35 â€Ş 39		2

3. Illustrating the data in tabular form .

Frequency Distribution for the Traffic Data

Number of autos per period	Number of periods
5-9	3
10 â€Ş 14	9
15 â€Ş 19	36
20 â€Ş 24	35
25 â€Ş 29	12
30 â€Ş 34	3
35 â€Ş 39	2
Total	**100**

In this example, the class marks of the traffic-count distribution are 7, 12, 17, â€ę, 32 and 37.

Example 10.3.1. The data below shows the mass of 40 students in a class. The measurement is to the nearest kg.

55	70	57	73	55	59	64	72
60	48	58	54	69	51	63	78
75	64	65	57	71	78	76	62
49	66	62	76	61	63	63	76
52	76	71	61	53	56	67	71

Construct a frequency table for the data using an appropriate scale.

Solution.

Step 1: Find the range.

The **range** of a set of numbers is the difference between the least number and the greatest number in the set

In this example, the greatest mass is 78 and the smallest mass is 48. The range of the masses is then 78 â€Ș 48 = 30. The scale of the frequency table must contain the range of masses.

Step 2: Find the intervals.

The intervals separate the scale into equal parts.

We could choose intervals of 5. We then begin the scale with 45 and end with 79.

Step 3: Draw the frequency table using the selected scale and intervals.

Mass (kg)	Frequency
45 â€Ș 49	2
50 â€Ș 54	4
55 â€Ș 59	7
60 â€Ș 64	10
65 â€Ș 69	4
70 â€Ș 74	6
75 â€Ș 79	7
Total	**40**

Frequency Distribution for continuous data.

The following data were obtained in a survey of the heights of 20 students in a sports club. Each height was measured to the nearest cm.

133	136	120	138	133	131	127	141	127	143
130	131	125	144	128	134	135	137	133	129

To form a frequency distribution of the heights of the 20 students, we group the information into classes or intervals.

The following are the different ways of writing the same set of intervals.

Height (cm)
$119.5 \leq b < 124.5$
$124.5 \leq b < 129.5$
$129.5 \leq b < 134.5$
$134.5 \leq b < 139.5$
$139.5 \leq b < 144.5$

Height (cm)
119.5 – 124.5
124.5 – 129.5
129.5 – 134.5
134.5 – 139.5
139.5 – 144.5

Height (cm)
120 – 124
125 – 129
130 – 134
135 – 139
140 – 144

The values 119.5, 124,5, 129.5, ... are called **class boundaries** or **interval boundaries.** The **upper class boundary** (u.c.b) of one interval is the **lower class boundary** (l.c.b) of the next interval.

10.3.2 Width of an interval

The width of an interval is the difference between the boundaries. Width of an interval = Upper class boundary - lower class boundary. Often intervals with equal widths are chosen, like in the above illustrations in which each width is 5cm.

To group the data, it helps to use a tally column, entering the numbers in the first row, and the second row etc.

The frequence distribution table would read as follows:

Height (cm)	Tally	Frequency
$119.5 \leq b < 124.5$		1
$124.5 \leq b < 129.5$		5
$129.5 \leq b < 134.5$		7
$134.5 \leq b < 139.5$		4
$139.5 \leq b < 144.5$		3
Total		20

10.4 Histograms

A histogram is usually used to present frequency distributions graphically. This is constructed by drawing rectangles over each class. The area of each rectangle should be proportional to its frequency.

Notes:

1. The vertical lines of a histogram should be the class boundaries.

2. If the smallest observation is far away from zero, then a break sign $\sim$ should be introduced in the horizontal axis.

3. The vertical axis will be labelled **frequency density**, where

$$\text{frequency density} = \frac{\text{frequency}}{\text{interval width}}.$$

Example 10.4.1. The table below shows the distribution of ages of passengers on a flight from Lusaka to Ndola.

Age, x years	$0 \leq x < 20$	$20 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 70$	$70 \leq$
Frequency	4	44	36	28	

Construct a histogram for the above data.

Solutions

We first begin by completing the frequency table.

Age, x years	Interval width	Frequency	Frequency density
$0 \le x < 20$	20	4	0.2
$20 \le x < 40$	20	44	2.2
$40 \le x < 50$	10	36	3.6
$50 \le x < 70$	20	28	1.4
$70 \le x < 100$	30	6	0.2

Age of passengers against frequency density.

The histogram will look like.

10.4.2 Modal class

The highest bar in the histogram represents the interval $40 \le x < 50$. This is the **modal class.** In a grouped frequency distribution, the modal class is the interval with the greatest frequency density. That is the interval represented by the highest bar in the histogram provided that the intervals have the same width.

10.5 Frequency Polygons

A grouped frequency distribution can be displayed as a frequency polygon. To construct a frequency polygon, for each interval, plot the frequency density against the midpoint value, where

$$\text{midpoint value} \;=\; \frac{1}{2}(l.c.b + u.c.b).$$

Then join the points with straight lines.

Example 10.5.1. Draw a frequency polygon to illustrate the following frequency distribution which gives the time taken by 31 students to finish a quiz test.

Time, t (min)	$25 \leq x < 30$	$30 \leq x < 35$	$35 \leq x < 40$	$40 \leq x < 50$	50
Frequency	4	12	8	4	

Solution

We begin by completing the frequency distribution table as follows.

Time, t (min)	Mid interval value	Interval width	Frequency	Frequenc
$25 \leq x < 30$	27.5	5	4	0
$30 \leq x < 35$	32.5	5	12	2
$35 \leq x < 40$	37.5	5	8	1
$40 \leq x < 50$	45	10	4	0
$50 \leq x < 65$	57.5	15	3	0

Below is the frequency polygon.

10.6 Cummulative Frequncy Curves (Ogives)

We recal that the frequency is the number of times an event occurs within a given scenario. **Cumulative frequency** is defined as the running total of frequencies. It is the sum of all the previous frequencies up to the current point. It is easily understandable through a Cumulative Frequency Table shown below

Marks	Frequency (number of students)	Cummulative Frequency
$0-5$	2	2
$5-10$	10	12
$10-15$	5	17
$15-20$	5	22

Cumulative Frequency is an important tool in Statistics to tabulate data in an organized manner. Whenever you wish to find out the popularity of a certain type of data, or the likelihood that a given event will fall within certain frequency distribution, a cumulative frequency table can be most useful. Say, for example, the Census department has collected data and wants to find out all residents in the city aged below 45. In this given case, a cumulative frequency table will be helpful.

Example 10.6.1. Draw a cummulative frequency curve for the following frequency distribution.

Height h (cm)	$150 \leq h < 155$	$155 \leq h < 160$	$160 \leq h < 165$	$165 \leq h$
Frequency	4	22	56	32

Solution As before, we begin by completing the frequency table.

Height h (cm)	< 150	< 155	< 160	< 165	< 170	< 175
Cummulative Frequency	0	4	26	82	114	119

Measures of central tendency are simply averages. The most commonly used are the arithmetic mean, mode and median.

10.7.1 The arithmetic mean (mean)

The mean of a set of observations $x_1, x_2, \dots, x_n$ is the sum of observations divided by the number of observations.

$$\text{mean} = \frac{\sum_{i=1}^{n} x_i}{n}.$$

It follows that the mean of the values $x_1, x_2, \dots, x_n$ with frequencies $f_1, f_2, \dots, f_n$ is

$$\text{mean} = \frac{1}{N}\sum_{i=1}^{n} x_i f_i, \text{ where } N = \sum_{i=1}^{n} f_i$$

Example 10.7.2. 1. Find the arithmetic mean for the scores 3, 8, 4, 6 and 7.

2. Find the mean of the following table showing the number of children per family.

No. Children (x)	Frequency (f)
0	5
1	8
2	12
3	18
4	9
5	9

10.7.3 Assumed mean

If a guess is made at the mean, say A, and the difference between the guess A and x_i is d_i, then

$$\bar{x} = A + \frac{\sum d_i}{n},$$

where $d_i = x_i - A$ or

$$\bar{x} = A + \frac{\sum f_i d_i}{N},$$

Example 10.7.4. Use a suitable guess to the arithmetic mean $\bar{x}$ of the numbers $485, 513, 498, 501, 512, 497$.

Solution Let $A = 500$. Then the values of d_i are $-15, 13, -2, 1, 12, -3$. Thus

$$\sum d_i = 6.$$

Consequently,

$$\bar{x} = A + \frac{\sum d_i}{n} = 500 + \frac{6}{6} = 501.$$

The value of this technique does not become apparent until large groups of data are handled.

For the grouped frequency distribution, we use the midinterval values to compute the mean.

10.8 The median (ungrouped data)

The median of a set off numbers is the middle value or arithmetic mean of the two middle values, if they are arranged in increasing order.

Example 10.8.1. Find the median of the following data.

1. $3, 0, 4, 8, -4, 6, 12, 5, 7$
2. $0, 0, 4, -3, 8, 12, 14, -3, 7, 26$

10.8.2 Median for grouped data

The median for grouped data is calculated using the formula

$$\text{Median } = L_1 + \left(\frac{\frac{1}{2}(n+1) - (\sum f)_1}{f_{\text{ median}}} \right) c,$$

where L_1 is the lower boundary of the class containing the median, n is the total frequency, $(\sum f)_1$ is the sum of all the frequencies in the classes below the median classs, f_{median} is the frequency of the class containing the median, c is the class width of the median class.

Note Classes do not all have to be the same width.

Example 10.8.3. The following frequency distribution is the summary of travel time to work for some employees

Time to travel to work (x)	Frequency (f)	Cummulative Frequency
1 – 10	8	8
11 – 20	14	22
21 – 30	12	34
31 – 40	9	43
41 – 50	7	50

Compute the median.

Solution $L_1 = 20.5$ $n = 50$, $(\sum f)_1 = 22$, $f_{\text{median}} = 12$ $c = 10$. Consequently

$$\text{Median} = L_1 + \left(\frac{\frac{1}{2}(n) - (\sum f)_1}{f_{\text{median}}}\right) c,$$

$$\text{Median} = 20.5 + \left(\frac{\frac{1}{2}(50) - 22}{12}\right) 10 = 20.5 + \frac{25 - 22}{12} 10 = 23$$

25 persons take less than 23 minutes to travel to work and another 25 persons take more than 23 minutes to travel to work.

10.9 The mode

The mode for a set of numbers is that value or values which occur most often. A set of numbers with one mode is called unimodal. A set of numbers with two modes is called bimodal.

10.9.1 Mode for grouped data

For groups data, the mode is defined by

$$\text{Mode} = L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2}\right) c,$$

where

L_1 is the lower boundary of the class containing the mode, c is the class width of the modal class, Δ_1 is the difference between the frequency of the modal class and the frequency of the previous lower class, Δ_2 is the difference between the frequency of the modal class and the frequency of thenext upper class.

or

$$\text{Mode} = L_1 + \left(\frac{f_{\text{mode}} - f_0}{2f_{\text{mode}} - f_0 - f_2}\right) c,$$

From the previous example,

$L_1 = 10.5$, $c = 10$, $\Delta_1 = 14 - 8 = 6$, $\Delta_2 = 14 - 12 = 2$.

Consequently,

$$\text{Mode} = 10.5 + \frac{6}{8} \times 10 = 18.$$

10.10 Measures of Dispersion

10.10.1 Range

The measure of dispersion which is easiest to understand and easiest to calculate is the **range.** Range is defined as: Range = Largest observation â€Ş Smallest observation.

10.10.2 Mean deviation

For n observation $x_1, x_2, \ldots, x_n$, the mean deviation about their mean $\bar{x}$ is given by

$$M.D(\bar{x}) = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}.$$

Example 10.10.3. Find the Mean deviation about the mean $(\bar{x})$ for the following data: $57, 64, 43, 67, 49, 59, 44, 47, 61, 59$.

The mean deviation about the median (M) is given by

$$M.D(M) = \frac{\sum_{i=1}^{n} |x_i - M|}{n}.$$

For grouped data, the above formulae become

$$M.D(\bar{x}) = \frac{\sum_{i=1}^{n} f_i |x_i - \bar{x}|}{n}$$

and

$$M.D(M) = \frac{\sum_{i=1}^{n} f_i |x_i - M|}{n}.$$

10.10.4 Standard Deviation

The standard deviation s of a set of values $\{x_1, x_2, \ldots, x_n\}$ is defined by

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}}.$$

For grouped frequency, the formula becomes

$$s = \sqrt{\frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})^2}{N}}.$$

Example 10.10.5. Find the standard deviation for the numbers $4, 5, 8, 9, 10$.

Solution $\bar{x} = 7.2$

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
4	-3.2	10.24
5	-2.2	4.84
8	0.8	0.64
9	1.8	3.24
10	2.8	7.84
Total 5		26.8

Hence,

$$s = \sqrt{\frac{26.8}{5}} = 2.315.$$

10.10.6 Variance

In many situations in statistcs, it is the square of the standard deviation, s^2, which is important. We call s^2 the variance.

From the previous example,

$$s^2 = 5.36.$$

NOTE: THESE LECTURE NOTES ON INTEGRAL CALCULUS ARE MEANT FOR MAT1110 (2020/2021) ACADEMIC YEAR ONLY.